1 Results

1. Gathered data
   (a) Structure of data from Silvics
      i. Angiosperm 79% (98/124) with masting data; Strong masting and weak masting split almost equally 52% (51/98); Gymnosperm 95% (61/64) with masting data; almost all species are strong masting 89% (54/61)
   (b) Number of traits for different hypotheses
      i. For angiosperm, we have data for:
         A. Predator satiation: seed weight, fruit size, oil content, seed dispersal mode, seed dormancy
         B. Pollination coupling: pollination mode, reproductive type
         C. Resource matching: drought tolerance, leaf longevity
      ii. For gymnosperm, we have the same data as for angiosperm, but we only have oil content for strong masting species and all the gymnosperm species are wind pollinated.

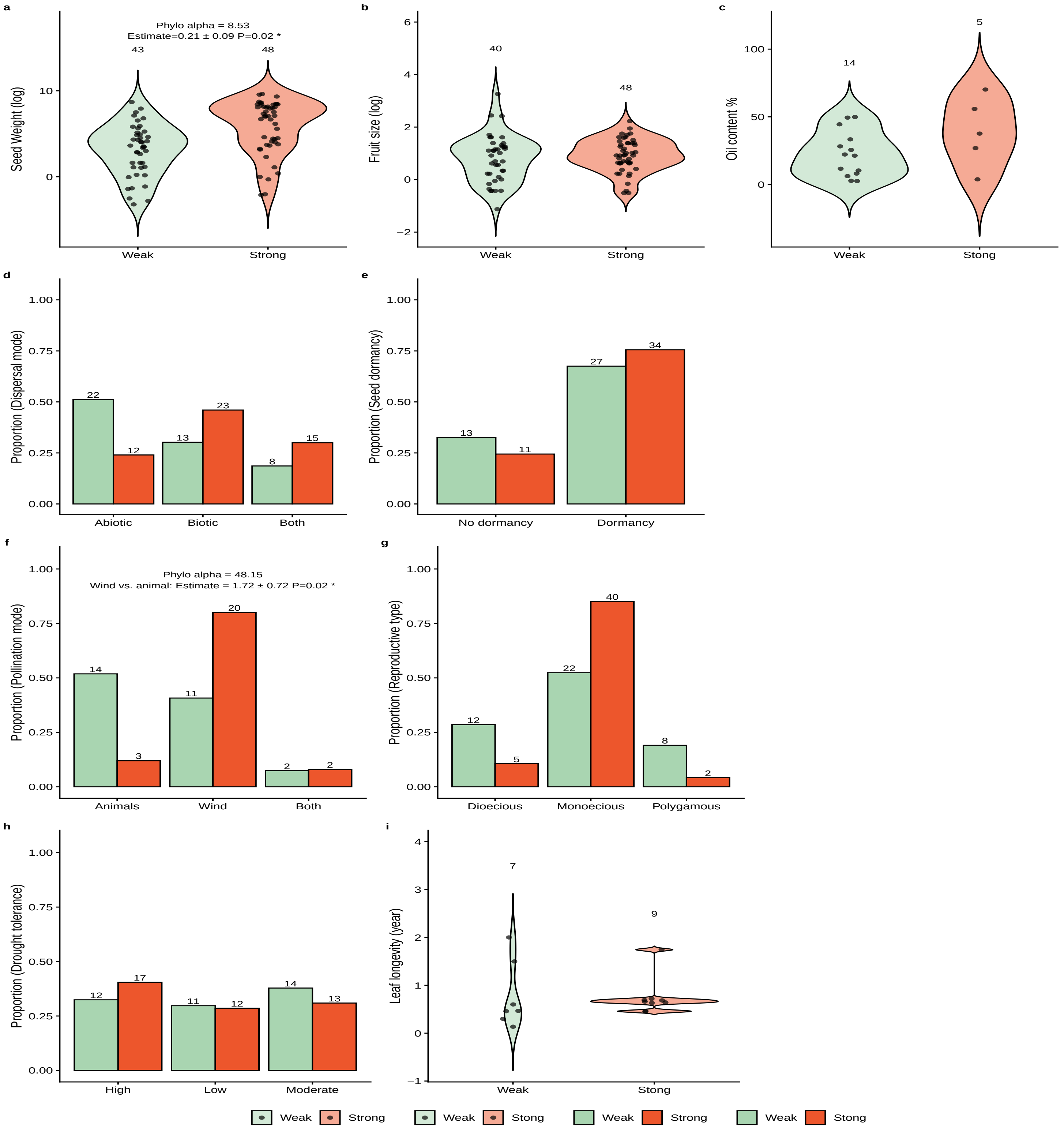

Figure 1: How plant traits related to different hypotheses differ in strong masting and weak masting for angiosperm species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (a-c, i) or proportional histograms (d-h) separated by species categorized as strong masting (red) or weak masting (green). We show model summaries only where significant (P-value < 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α. Full model results are show in Table S1.

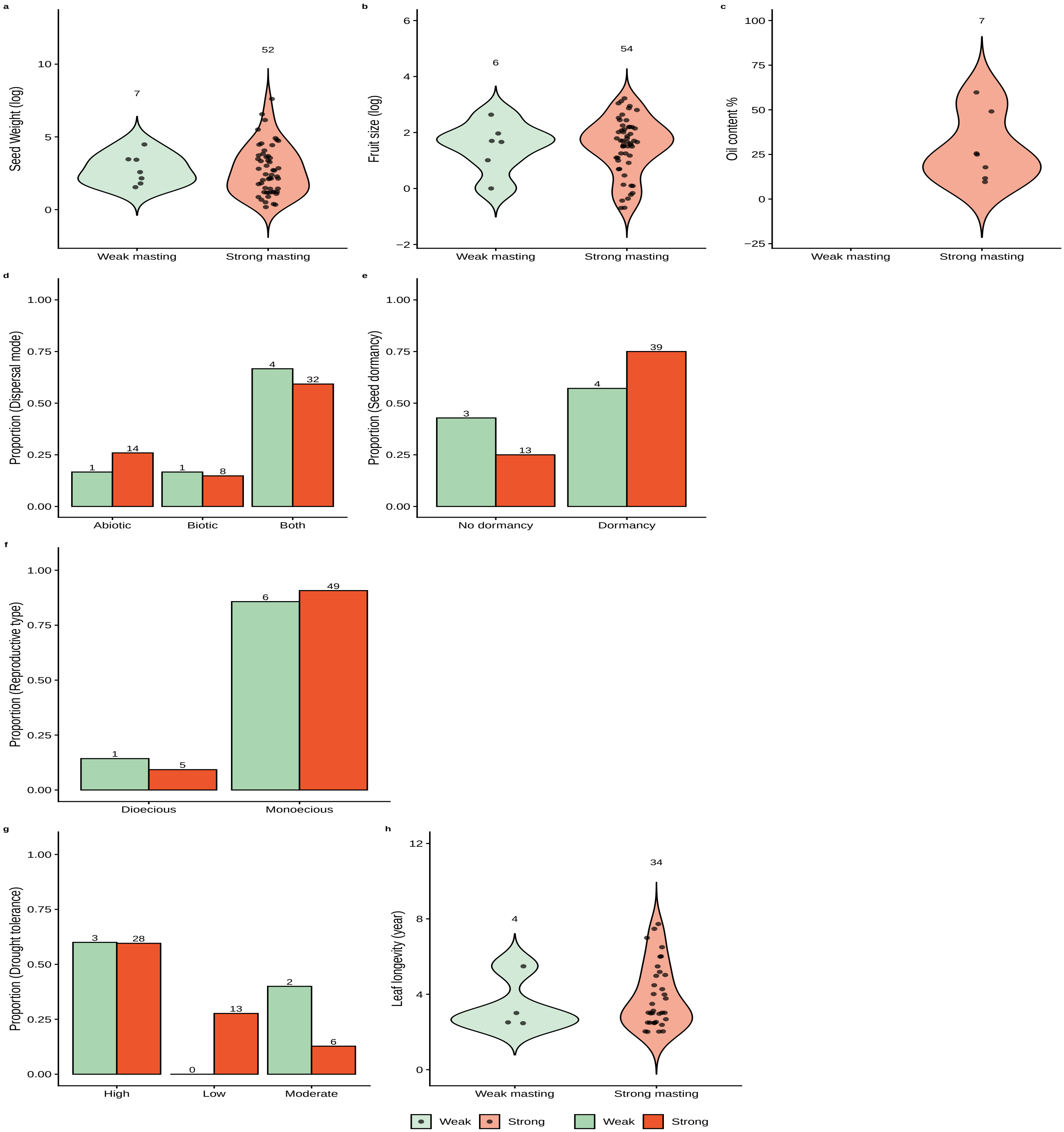

Figure 2: How plant traits related to different hypotheses differ in strong masting and weak masting for gymnosperm species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (a-c, i) or proportional histograms (d-h) separated by species categorized as strong masting (red) or weak masting (green). We show model summaries only where significant (P-value $<$ 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α. Full model results are show in Table S2.

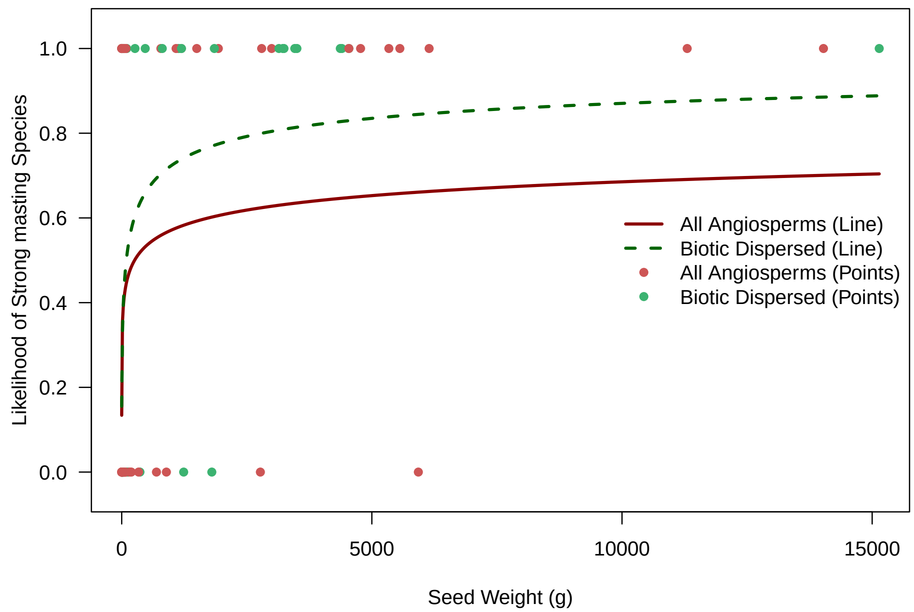

Figure 3: Predicted likelihood of masting as a function of seed weight for all angiosperm species (solid red line) and the biotic-dispersed subgroup (dashed green line), estimated using phylogenetic logistic regression. Individual species observations are plotted as binary data points at 0 (weak masting) and 1 (strong masting).

2. Results for each hypothesis
   (a) Predation satiation: seed weight only with a small effect and only for angiosperm
      i. Only seed weight is related to masting with larger seeds increasing the probability of being a strong masting species (Figure 1). For angiosperms only, tripling of seed weight (from a median seed weight of 90.9g to 272.7g) is predicted to increase the likelihood of strong masting from 0.45 to 0.51 (Figure 3). And this increase is stronger in biotic-dispersed seed group, tripling of seed weight (from a median seed weight of 1087.15g to 3261.45g) is predicted to increase the likelihood of strong masting from 0.72 to 0.80 (Table S3; Figure 3).
   (b) Pollination satiation: wind pollination for angiosperm
      i. We found wind pollination is related to strong masting for angiosperm species, likelihood for wind-pollinated species is predicted to increase the likelihood of strong masting to 0.62, whereas the likelihood is 0.20 for animal-pollinated species. Gymnosperm are all wind pollinated, and gymnosperm has a greater portion of strong masting species.
   (c) Resource matching: nothing significant

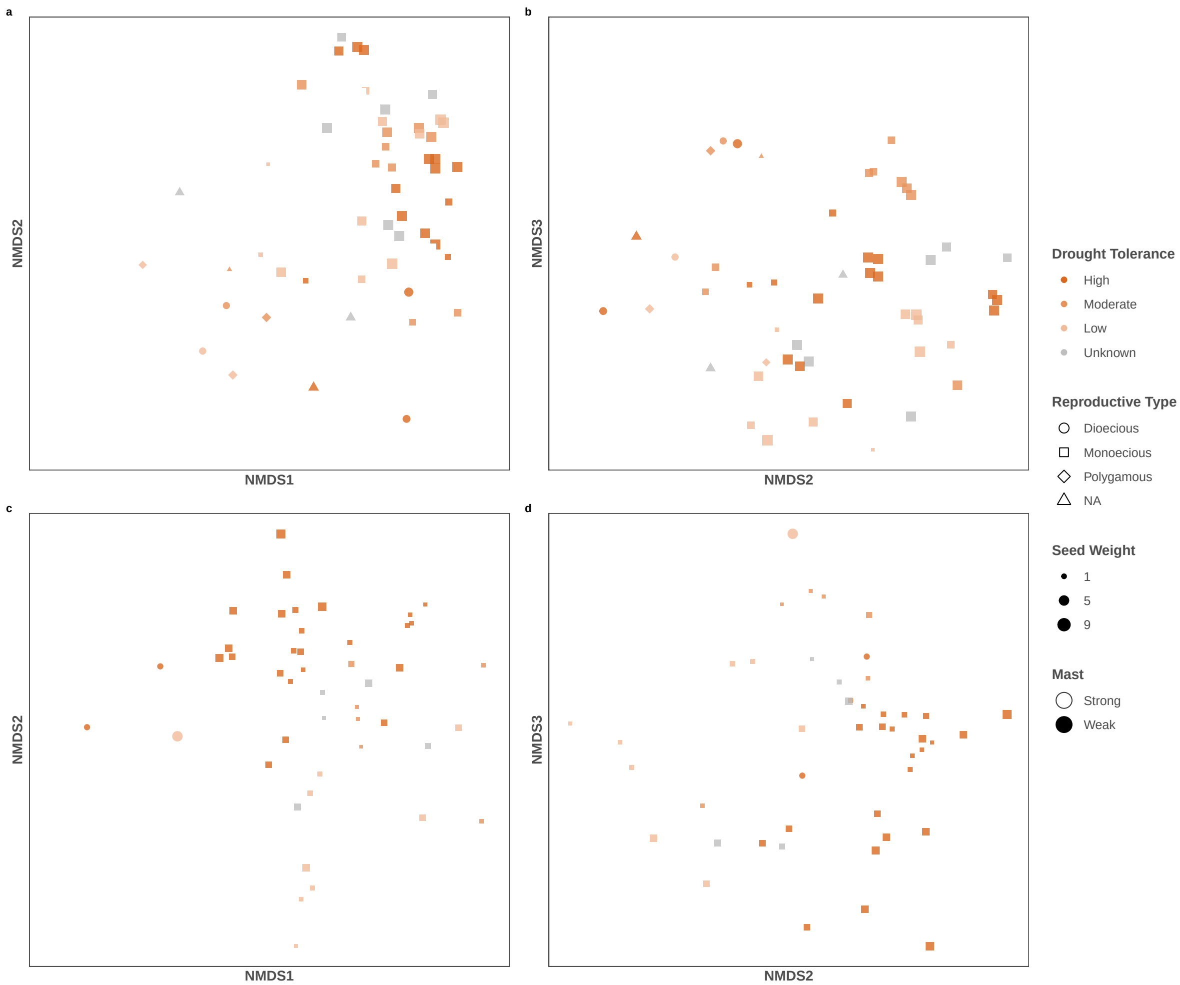

Figure 4: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and gymnosperm species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by strong masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) gymnosperm NMDS1–NMDS2, and (d) gymnosperm NMDS2–NMDS3.

3. Multidimension-NMDS
   (a) No strong trend when considering single trait, maybe because it happens in multidimensional space
      i. Masting is not aligned well along any axis in the NMDS analysis.
      ii. We do not have evidence that the combination of traits for any of the hypotheses is driving the pattern of masting.

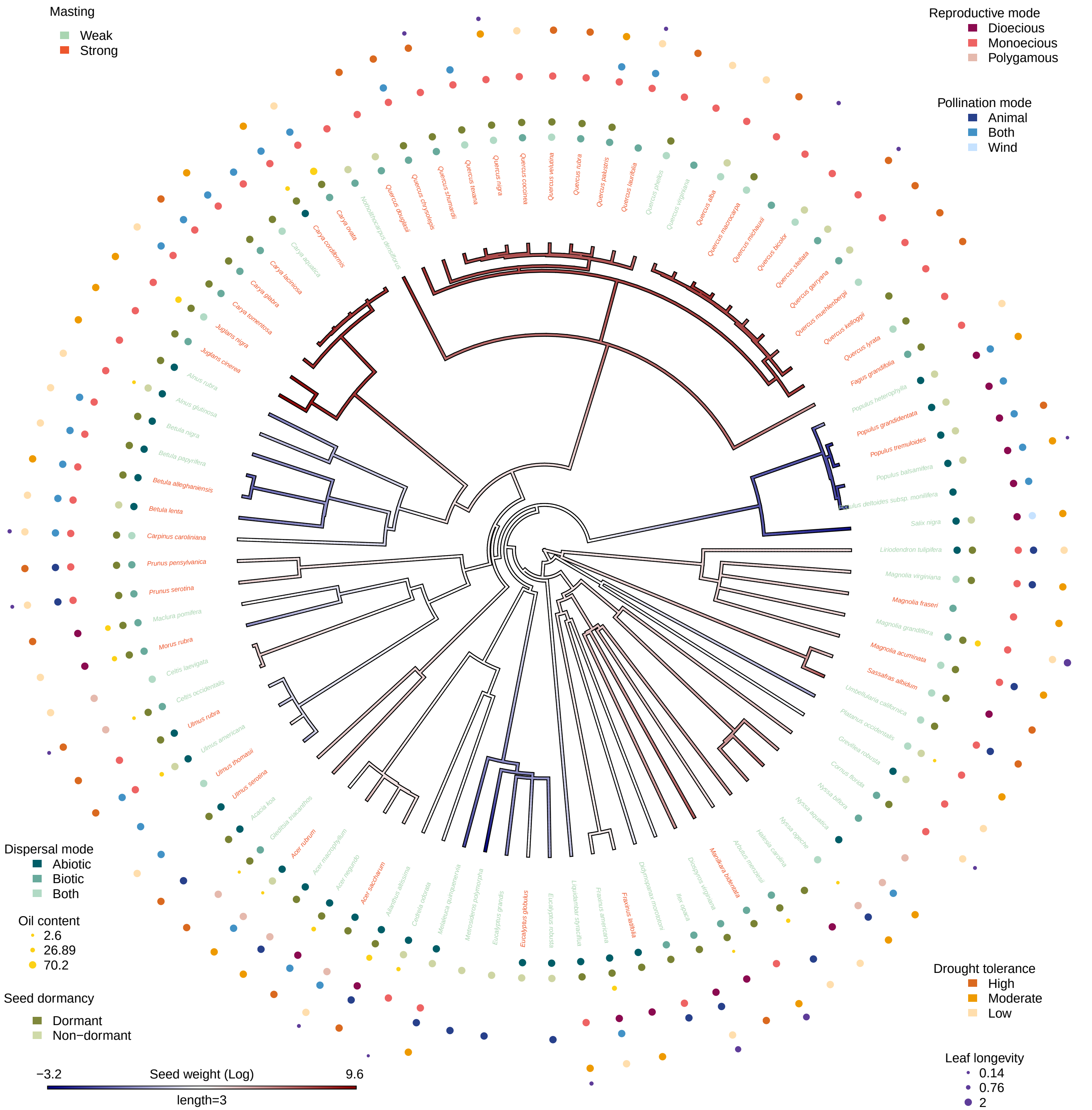

Figure 5: Phylogenetic tree for all angioperm species included in the analysis. The color scheme of the branch is the trait value for seed weight (log). The color tip is colored by strong masting or not. The other traits are labeled after the tip labels with different color and size. We also show the predictions of relationship between traits and strong masting. We calculated the Pagel's λ for continuous traits as well as multilevel categorical traits, and calculated D-statistic for binary traits.

4. Phylogenetic patterns

   (a) Pagel's λ for continuous traits and multilevel categorical traits in Table S4, D-statistic for binary traits in Table S5.

      i. Strong masting shows a moderate phylogenetic signal in both gymnosperms (estimated D = 0.51) and angiosperms (estimated D = 0.58) (Table S5).

      ii. Seed weight has strong phylogenetic signal for both angiosperm ($\lambda = 0.95$) and gymnosperm ($\lambda = 0.97$). We can see the similarity in certain group in Figure 5 it's relatively reliable estimate. Dispersal mode, leaf longevity, pollination mode and reproductive type also showed high phylogenetic signal in Table S4, but be cautious that λ for categorical traits might not be accurate.

      iii. Be cautious that λ might not be comparable among traits, not only because the way λ is calculated for continuous traits and categorical traits are different, but also because λ might depend on the data structure when sampling is uneven, take leaf longevity as an example in Figure 5,leaf longevity data is sparse and nested in *Quercus*. (5/16).

A Supplementary Material

Trait	Type	Predictor	N	Estimate	SE	P	Phylo α	Half-life
Seed weight (log)	Numeric	Seed weight (log)	91	0.21	0.09	0.02	8.53	0.08
Fruit size (log)	Numeric	Fruit size (log)	88	0.08	0.31	0.79	19.04	0.04
Oil content (%)	Numeric	Oil content (%)	19	0.04	0.03	0.20	19.08	0.04
Seed dormancy	Categorical	Dormant vs. non-dormant	85	0.05	0.51	0.92	12.19	0.06
Dispersal mode	Categorical	Biotic vs. abiotic	93	0.53	0.53	0.32	8.11	0.09
Dispersal mode	Categorical	Both vs. abiotic	93	0.53	0.57	0.35	8.11	0.09
Pollination mode	Categorical	Wind vs. animal	52	1.72	0.72	0.02	48.15	0.01
Pollination mode	Categorical	Wind vs. animal and animals	52	1.18	1.15	0.31	48.15	0.01
Reproductive type	Categorical	Monoecious vs. dioecious	89	0.76	0.68	0.26	13.73	0.05
Reproductive type	Categorical	Monoecious vs. polygamous	89	-0.29	0.96	0.76	13.73	0.05
Drought tolerance	Categorical	Low vs. high drought tolerance	79	0.15	0.47	0.76	11.35	0.06
Drought tolerance	Categorical	Moderate vs. high drought tolerance	79	-0.15	0.45	0.74	11.35	0.06
Leaf longevity (years)	Numeric	Leaf longevity (years)	16	0.26	0.95	0.78	0.90	0.77

Table S1: Phylogenetic generalized linear model results for angiosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical (Type column), where we report (Estimate column) the the difference in log-odds between a given level and its designated baseline reference level, or continuous, where we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), Phylo α, which represents the the phylogenetic signal in the residuals estimated for each trait, it has an upper bound of 54. If α reached 54, then the phylogenetic correlation is estimated to be negligible. We also calculated half-life ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values greater than 1 also indicate weak phylogenetic signal in the residuals.

Trait	Type	Predictor	N	Estimate	SE	P	Phylo α	Half-life
Seed weight (log)	Numeric	Seed weight (log)	59	0.01	0.24	0.98	54.52	0.01
Fruit size (log)	Numeric	Fruit size (log)	60	0.03	0.36	0.94	54.14	0.01
Seed dispersal	Categorical	Biotic vs. abiotic	60	-0.83	0.81	0.30	1.95	0.36
Seed dispersal	Categorical	Both vs. abiotic	60	-1.11	0.67	0.10	1.95	0.36
Seed dormancy	Categorical	Dormant vs. non-dormant	59	0.69	0.69	0.32	48.04	0.01
Reproductive type	Categorical	Monoecious vs. dioecious	61	-1.13	1.01	0.26	0.89	0.78
Drought tolerance	Categorical	Low vs. high drought tolerance	52	11.36	98.16	0.91	0.02	34.66
Drought tolerance	Categorical	Moderate vs. high drought tolerance	52	-0.29	0.60	0.63	0.02	34.66
Leaf longevity (years)	Numeric	Leaf longevity (years)	38	0.15	0.35	0.67	54.50	0.01

Table S2: Phylogenetic generalized linear model results for gymnosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical (Type column), where we report (Estimate column) the the difference in log-odds between a given level and its designated baseline reference level, or continuous, where we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), Phylo α, which represents the the phylogenetic signal in the residuals estimated for each trait, it has an upper bound of 54. If α reached 54, then the phylogenetic correlation is estimated to be negligible. We also calculated half-life ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values greater than 1 also indicate weak phylogenetic signal in the residuals.

Trait	Group	Predictor	N	Estimate	SE	P	Phylo α	Half-life
Seed weight (log)	Biotic and both	Seed weight (log)	55	0.41	0.16	0.01	54.57	0.01
Seed weight (log)	Abiotic	Seed weight (log)	36	0.12	0.13	0.36	50.33	0.01

Table S3: Phylogenetic generalized linear model results for only seed weight for biotic-dispersed subgroup (both group included) and abiotic-dsipersed subgroup. We report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor), Phylo α, which represents the the phylogenetic signal in the residuals estimated for each trait, it has an upper bound of 54. If α reached 54, then the phylogenetic correlation is estimated to be negligible. We also calculated half-life ($\log(2)/\alpha$), which represents the time required for the influence of ancestral states to decrease by half, where values greater than 1 also indicate weak phylogenetic signal in the residuals.

Group	Trait	λ
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	0.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.98
Gymnosperm	Fruit size (Log)	0.91
Gymnosperm	Oil content %	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	0.00
Gymnosperm	Reproductive type	1.00
Gymnosperm	Drought tolerance	0.00

Table S4: Phylogenetic Signal (Pagel's λ) for continous traits and multi-level categorical traits. $\lambda = 0$ indicates no phylogenetic signal, trait districuted independent of the phylogeny; $0 < \lambda < 1$ means moderate phylogenetic signal, trait is partially structured by the phylogeny; $\lambda = 1$ indicates strong phylogenetic signal, trait follows Brownian motion.

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Seed dormancy	0.25	0	0.16
Angiosperm	Masting	0.58	0	0.01
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0

Table S5: Phylogenetic Signal (D-statistic) for the phylogenetic signal of binary traits, with probabilities calculated under a random phylogenetic structure and under a Brownian motion phylogenetic structure. $D = 1$ indicates the trait is randomly distributed across the phylogeny; $D = 0$ indicates the trait follows a Brownian motion model; $0 < D < 1$ indivates moderate phylogenetic signal; $D < 0$ indicates the trait is more clumped than expected under Brownian motion; and $D > 1$ indicates overdispersion. P random is the p-value for the null hypothesis that the trait is randomly distributed across the phylogeny (no phylogenetic signal), and P Brownian is the p-value for the null hypothesis that the trait follows a Brownian motion model (strong phylogenetic signal). A low p-value indicates rejection of the corresponding null hypothesis.